

CHAPTER 2

What Makes a Good Business Experiment?

Our success at Amazon is a function of how many experiments we do per year, per month, per week, per day.

—JEFF BEZOS, CEO, AMAZON

In 2016, Jeff Bezos gave shareholders a rare insight into Amazon’s innovation engine. In his annual letter, he explained: “One area where I think we are especially distinctive is failure. I believe we are the best place in the world to fail (we have plenty of practice!), and failure and invention are inseparable twins. To invent you have to experiment, and if you know in advance that it’s going to work, it’s not an experiment. Most large organizations embrace the idea of invention, but are not willing to suffer the string of failed experiments necessary to get there.”

Bezos didn’t stop there. For him, the business logic for tolerating, even inviting, failure came from the outsized economic returns of winning. He explained why experiments have been so important to Amazon’s growth model:

Outsized returns often come from betting against conventional wisdom, and conventional wisdom is usually right. Given a ten percent chance of a 100 times payoff, you should take that bet every time. But you’re still going to be wrong nine times out of ten. We all know that if you swing for the fences, you’re going to strike out a lot, but you’re also going to hit some home runs. The difference between baseball and business, however, is that baseball has a truncated outcome distribution. When you swing, no matter how well you connect with the ball, the most runs you can get is four. In business, every once in a while, when you step up to the plate, you can score 1,000 runs. This long-tailed distribution of returns is why it’s important to be bold.¹

Business experimentation has become part and parcel of how Amazon makes decisions, through a process Bezos calls “the unnatural thing of trying to disconfirm our beliefs.” After all, humans strongly prefer evidence that confirms their preexisting beliefs. But such confirmation bias gets in the way of making decisions about innovation, an arena in which most ideas don’t work. So it should not have come as a surprise when, partially in an effort

to kick its experiments up a notch, Amazon acquired Whole Foods about fourteen months after Bezos wrote the letter. Industry observers felt that its physical supermarkets could become laboratories for radical experiments. Correspondingly, share prices of competing grocery chains plummeted after the announcement.² Amazon's reputation for fearless innovation was fueled by such radical business experiments—so-called *big swings*—and, equally important, the tens of thousands of smaller and disciplined experiments that have led to a highly optimized user experience in its web store.

If the case for business experimentation is so compelling, then why don't more companies conduct rigorous tests of their risky overhauls and expensive innovation proposals in order to make better decisions? Why do executives rely on hierarchy, persuasion, or PowerPoints instead of demanding that teams present experimental evidence before making business decisions? Clearly, there are cultural obstacles that inhibit experimentation. It's also true that managers often misuse the term, saying "*We experiment*" in lieu of "*We are trying something new,*" but without putting enough thought into the discipline and rigor needed to get useful test results. In the most egregious cases, projects or business initiatives become "experiments" after they are finished, in an effort to excuse poor execution.

But I've also found that many organizations are reluctant to fund good business experiments and have considerable difficulty executing them. Although the process of experimentation seems straightforward, it is surprisingly hard in practice, owing to myriad organizational, management, and technical challenges. Moreover, most tests of new business initiatives are too informal. They are not based on proven scientific and statistical methods, and so executives end up misinterpreting statistical noise as causation—and make bad decisions. In [chapter 1](#), we looked at the operational drivers that accelerate learning. But going faster doesn't guarantee that decisions are made better; companies may just make more decisions, more quickly. In this chapter, we will see *how* companies can run good experiments by systematically following a clear set of principles.³

As we saw earlier, in an ideal experiment, testers separate an independent variable (the presumed cause) from a dependent variable (the observed effect) while holding all other potential causes constant. They then manipulate the former to study changes in the latter. The manipulation, followed by careful observation and analysis, yields insight into the relationships between cause and effect, which ideally can be applied and tested in other settings. To obtain that kind of learning—and ensure that each experiment yields better decisions—companies should ask themselves seven important questions: (1) Does the experiment have a testable hypothesis? (2) Have stakeholders made a commitment to

abide by the results? (3) Is the experiment doable? (4) How can we ensure reliable results? (5) Do we understand cause and effect? (6) Have we gotten the most value out of the experiment? And finally, (7) Are experiments *really* driving our decisions? (See [table 2-1.](#)) Although some of the questions seem obvious, many companies conduct tests without fully addressing them.

Question 1: Does the Experiment Have a Testable Hypothesis?

Companies should conduct experiments if they are the only practical way to answer specific questions about proposed management actions—and of course, if the answer isn’t already obvious.⁴ Consider Kohl’s, the large retailer, which in 2013 was looking for ways to decrease its operating costs. One suggestion was to open stores an hour later on Monday through Saturday. Company executives were split on the matter. Some argued that reducing the stores’ hours would result in a significant drop in sales; others claimed that the impact on sales would be minimal. The only way to settle the debate with any certainty was to conduct a controlled experiment (see the sidebar, “[What Is a Controlled Experiment?](#)”). A test involving one hundred of the company’s stores showed that the delayed opening would not result in any meaningful sales decline.

In determining whether an experiment is needed, managers must first figure out exactly what they want to learn and measure. Only then can they decide if testing is the best approach to achieve their answer and, if so, what the scope of the experiment should be. In the case of Kohl’s, the hypothesis to be tested was straightforward: Opening stores an hour later to reduce operating costs will not lead to a significant drop in sales. This is referred to as the *null hypothesis*, which is the default statement when there is no measureable change. The null hypothesis is generally assumed to be reliable until empirical evidence suggests otherwise. And when the result of an experiment is *statistically significant*, it simply means that the observed (sample average) drop is unlikely to be the result of chance. Similarly, if the result is *not* statistically significant, it does not prove that a treatment had no impact; it simply means that either the observed change or the sample size is not large enough to support a finding (and the related decision) with sufficient confidence.

TABLE 2-1

Questions for running good business experiments

- | | |
|----------------|--|
| 1. Hypothesis | <ul style="list-style-type: none"> • Is the hypothesis rooted in observations, insights, or data? • Does the experiment focus on a testable management action under consideration? • Does it have measurable variables, and can it be shown to be false? • What do people hope to learn from the experiment? |
| 2. Buy-in | <ul style="list-style-type: none"> • What specific changes would be made on the basis of the results? • How will the organization ensure that the results aren't ignored? • How does the experiment fit into the organization's overall learning agenda and strategic priorities? |
| 3. Feasibility | <ul style="list-style-type: none"> • Does the experiment have a testable prediction? • What is the required sample size? <i>Note:</i> The sample size will depend on the expected effect (for example, a 5 percent increase in sales). • Can the organization feasibly conduct the experiment at the test locations for the required duration? |
| 4. Reliability | <ul style="list-style-type: none"> • What measures will be used to account for systemic bias, whether it's conscious or unconscious? • Do the characteristics of the control group match those of the test group? • Can the experiment be conducted in either "blind" or "double-blind" fashion? • Have any remaining biases been eliminated through statistical analyses or other techniques? • Would others conducting the same test obtain similar results? |
| 5. Causality | <ul style="list-style-type: none"> • Did we capture all variables that might influence our metrics? • Can we link specific interventions to the observed effect? • What is the strength of the evidence? Correlations are merely suggestive of causality. • Are we comfortable taking action without evidence of causality? |
| 6. Value | <ul style="list-style-type: none"> • Has the organization considered a targeted rollout—that is, one that takes into account a proposed initiative's effect on different customers, markets, and segments—to concentrate investments in areas when the potential payback is the highest? • Has the organization implemented only the components of an initiative with the highest return on investment? • Does the organization have a better understanding of what variables are causing what effects? |
| 7. Decisions | <ul style="list-style-type: none"> • Do we acknowledge that not every business decision can or should be resolved by experiments but that everything that can be tested should be tested? • Are we using experimental evidence to add transparency to our decision-making process? |

What Is a Controlled Experiment?

In a controlled experiment, customers are randomly exposed to one of multiple variants. One is the *control*—the current practice (for example, the way an online ad is presented). The other variants are the *treatments*—the current practice with some modifications (for example, the same online ad, but with different-colored headlines). The experiment exposes *samples* of

users to the control and the treatment, and computes the sample averages for the metric of interest (e.g., revenue). The *treatment effect* is the difference between the sample averages.

It's important here to note an important tenet of the scientific method—that experiments can refute, but not prove, a hypothesis. This important tenet of the scientific method is neatly worded by Albert Einstein: “No amount of experimentation can ever prove me right; a single experiment can prove me wrong.”⁵ A new fact is thus established if repeated, rigorous experiments fail to refute the null hypothesis. For Kohl's, management's hypothesis—“Opening one hour later has no impact on sales”—was not proven by their experiment(s); it simply wasn't rejected.⁶

All too often, though, companies lack the discipline to hone their hypotheses, leading to tests that are inefficient, unnecessarily costly, or, worse, ineffective in answering the questions at hand. A weak hypothesis—for example, “We can extend our brand upmarket”—doesn't present a specific independent variable to test on a specific dependent variable and cannot yield measurable outcomes. Thus, it is difficult either to support or to reject it. A good hypothesis helps delineate those variables and suggests metrics ([table 2-2](#)). Consider what the physicist William Thomson, better known as Lord Kelvin and father of the first two laws of thermodynamics, observed about science and knowledge: “When you can measure what you are speaking about and express it in numbers, you know something about it; but when you cannot measure it, when you cannot express it in numbers, your knowledge is of meager and unsatisfactory kind: it may be the beginning of knowledge, but you have scarcely, in your thoughts, advanced the stage of science, whatever the matter may be.”⁷

If the science of management is about building and organizing knowledge through testable explanations and predictions, then perhaps the debatable maxim “What gets measured, gets done” ought to be replaced by “What gets measured, gets explored (and hopefully understood).”

TABLE 2-2

What is a strong business hypothesis?

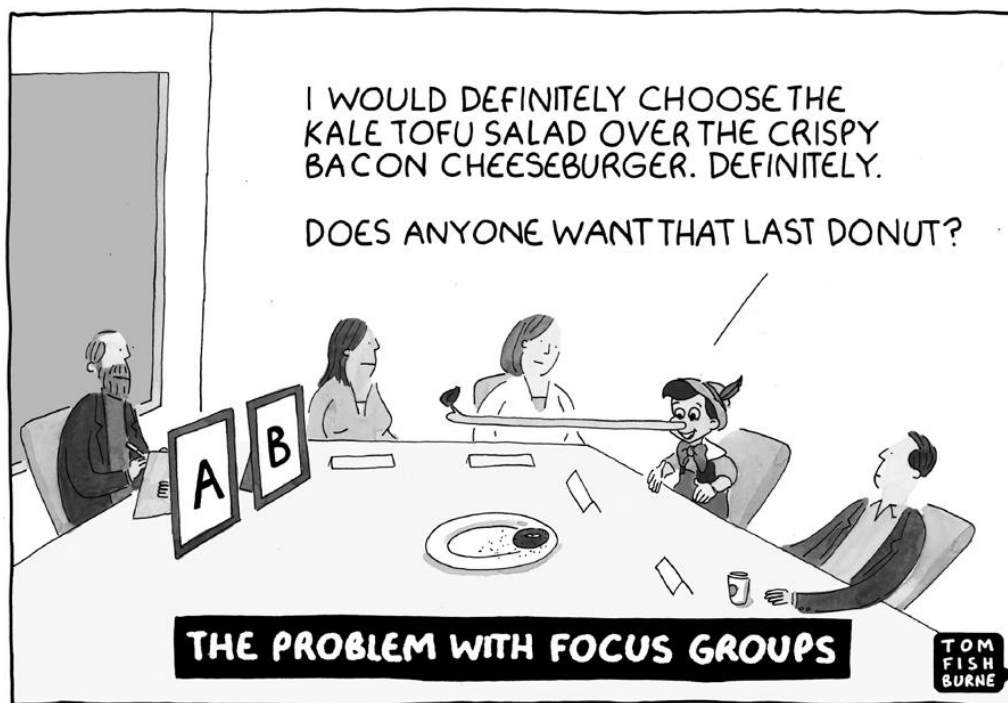
	Strong hypothesis	Weak hypothesis
Source	Qualitative research, customer insights, problems, observations, data mining, competitors	Guesses not rooted in observations or facts
Variables	Identifies possible cause and effect	Possible cause or effect unknown
Prediction	Can be shown to be false	Difficult to disprove, vague
Measurement	Quantifiable metrics	Qualitative outcomes
Verification	Experiment (with hypothesis) can be replicated	Difficult to repeat experiment
Motivation	Clear impact on business outcomes	Link between metric and business impact unclear
Example	“Opening our stores one hour later has no impact on daily sales revenue.”	“We can extend our brand upmarket.”

Good hypotheses often come from customer insights: qualitative research (e.g., focus groups, usability labs), analytics (e.g., patterns found in customer support data), or even serendipity. Consider what happened at the financial software company Intuit when an engineer noticed that about 50 percent of prospective customers tried the company’s small business product twenty minutes before they had to make payroll.⁸ The problem was that all payroll companies took hours or even days to approve new customers before the first employee could be paid. Wouldn’t potential customers be very pleased if they could make payroll before the long approval process was done? To make sure that there was a genuine need, the engineer and product manager ran a usability study. The result: none of the twenty participants were interested in a fast payroll solution. But instead of shelving the idea, Intuit modified its web page within twenty-four hours and ran a simple experiment that offered two versions of the software—one with the option to click on “Pay employees first” and another one with “Do setup first.” (When users clicked on the “Pay employees first” option, they got a message that the feature wasn’t ready.) Contrary to the usability

test results, the experiment revealed that 58 percent of new users picked the faster payroll option. Ultimately, the feature became hugely popular, lifted the software's conversion rate by 14 percent, and generated millions in additional revenue.

The team also discovered that testing customers' actual behavior is more important than trusting what they say they will do. It's not unusual to run into this saying-doing gap in customer focus groups. That's what the Dutch technology company Philips realized when it conducted a focus group of teenagers to assess their color preferences for a new boom box. During the session, most teenagers selected "yellow" as their preferred color. After the session, the teenagers received a boom box as a reward for their participation and were offered a choice of two colors: yellow and black. Most participants selected "black," even though they had chosen "yellow" when their preference was posed as a hypothetical question.⁹ When it comes to behavior, it's usually better to trust the experiment.

The saying-doing gap in focus groups



In many situations, executives need to go beyond the direct effects of an initiative and investigate its ancillary effects. For example, when Family Dollar wanted to determine whether to invest in refrigeration units so that it could sell eggs, milk, and other perishables, it discovered that a side effect—an increase in the sales of traditional dry goods to the

additional customers drawn to the stores by the refrigerated items—would actually have a bigger impact on profits. Ancillary effects can also be negative. A few years ago, Wawa, the convenience store chain operating in the mid-Atlantic United States, wanted to introduce a flatbread breakfast item that had done well in spot tests. But the initiative was killed before the launch when a rigorous experiment—complete with test and control groups followed by regression analyses—showed that the new product would likely cannibalize other more profitable items.¹⁰

Question 2: Have Stakeholders Made a Commitment to Abide by the Results?

Before conducting any test, stakeholders must agree how they'll proceed once the results are in. They should promise to weigh all the findings, rather than cherry-picking data that supports a particular point of view. Perhaps most importantly, they must be willing to walk away from a project if it's not supported by the data. But that's easier said than done.

When Kohl's was considering adding a new product category—furniture—many executives were tremendously enthusiastic, anticipating significant additional revenue. A test at seventy stores over six months, however, showed a net decrease in revenue. Products that now had less floor space (to make room for the furniture) experienced a drop in sales, and Kohl's was actually losing customers overall. Those negative results were a huge disappointment for those who had advocated for the initiative, but the program was nevertheless scrapped. The Kohl's example highlights the fact that experiments are often needed to perform objective assessments of initiatives backed by people with organizational clout. Of course, there might be good reasons for rolling out an initiative even when the anticipated benefits are not supported by the data—for example, a program that experiments have shown will not substantially boost sales might still be necessary to build customer loyalty. But if the proposed initiative is a done deal, why go through the time and expense of conducting a test? In such cases, it's best to call programs for what they are—a rollout, commitment, or implementation. (A possible litmus test is reversibility: If a new program cannot be easily reversed, it certainly fails one of the most basic attributes of an experiment.)

To ensure an organization abides by the results, there must be a process that ensures test results aren't ignored, even when they contradict the assumptions or intuition of top executives. At Publix Super Markets, a chain in the southeastern United States, virtually all

large retail projects, especially those requiring considerable capital expenditures, must undergo formal experiments to receive a green light. Proposals go through a filtering process in which the first step is for finance to perform an analysis to determine if an experiment is worth conducting. For projects that make the cut, analytics professionals develop test designs and submit them to a committee that includes the vice president of finance. An internal test group then conducts and oversees the experiments approved by the committee. Finance will approve significant expenditures only for proposed initiatives that have adhered to this process and whose experiment results are positive. “Projects get reviewed and approved much more quickly—and with less scrutiny—when they have our test results to back them,” according to Frank Maggio, the senior manager of business analysis at Publix.¹¹

When constructing and implementing such a filtering process, it is important to remember that experiments should be part of a learning agenda that supports a firm’s organizational priorities. At Petco, the pet supplies retailer, each test request must address how that particular experiment would contribute to the company’s overall strategy to become more innovative. In the past, the company performed about one hundred tests a year, but that number has since been trimmed to seventy-five. Many test requests have been denied because the company has done a similar test in the past; others are rejected because the changes under consideration are not radical enough to justify the expense of testing (for example, a price increase of a single item from \$2.79 to \$2.89). As John Rhoades, the company’s former director of retail analytics, noted, “We want to test things that will grow the business. We want to try new concepts or new ideas.”¹²

Question 3: Is the Experiment Doable?

As we have previously seen, experiments must have testable predictions. But the *causal density* of the business environment—that is, the complexity of the variables and their interactions—can make it extremely difficult to determine cause-and-effect relationships. Learning from a business experiment is not necessarily as easy as isolating an independent variable, manipulating it, and observing changes in the dependent variable. Environments are constantly changing, and the potential causes of business outcomes are often uncertain or unknown, so linkages between them are frequently complex and poorly understood.

Consider a hypothetical retail chain that owns ten thousand convenience stores, eight thousand of which are named QwikMart and two thousand, FastMart. The QwikMart stores have been averaging \$1 million in annual sales and the FastMart stores \$1.1 million. A

senior executive asks a seemingly simple question: Would changing the name of the QwikMart stores to FastMart lead to an increase in revenue of \$800 million? Obviously, numerous factors affect store sales, including the physical size of the store, the number of people who live within a certain radius and their average incomes, the number of hours the store is open per week, the experience of the store manager, the number of nearby competitors, and so on. But the executive is interested in just one variable: the stores' names.¹³

The obvious solution is to conduct an experiment by changing the name of a handful of QwikMart stores (say, ten) to see what happens. But even determining the effect of the name change on those stores turns out to be tricky, because many other variables may have changed at the same time. For example, the weather was very bad at four of the locations, a manager was replaced in one, a large residential building opened near another, and a competitor started an aggressive advertising promotion near yet another. Unless the company can isolate the effect of the name change from those and other variables, the executive won't know for sure whether the name change has helped (or hurt) business.

To deal with environments of high causal density, companies need to consider whether it's feasible to use a sample large enough to average out the effects of all variables except those being studied. Unfortunately, that type of experiment is not always doable. The cost of a test involving an adequate sample size might be prohibitive, or the change in operations could be too disruptive. In such instances, as we'll see later, executives can sometimes employ sophisticated analytical techniques, some involving big data, to increase the statistical validity of their results. That said, it should be noted that managers often mistakenly assume that a larger sample will automatically lead to better data. Indeed, an experiment can involve a lot of observations, but if they are highly clustered, or correlated to one another, then the true sample size might actually be quite small. When a company uses a distributor instead of selling directly to customers, for example, that distribution point could easily lead to correlations between customer data.

The required sample size depends in large part on the magnitude of the expected effect. If a company expects the cause (for example, a change in store name) to have a large effect (a substantial increase in sales), the sample can be smaller. If the expected effect is small, the sample must be larger. This might seem counterintuitive, but think of it this way: the smaller the expected effect, the greater the number of observations that are required to distinguish it from the surrounding noise with the desired statistical confidence. Selecting the right sample size does more than ensure that the results will be statistically valid; it can

also enable a company to decrease testing costs and increase innovation. Readily available software tools can help companies choose the optimal sample size.

Question 4: How Can We Ensure Reliable Results?

The previous section described the basics for conducting an ideal experiment. However, the truth is that companies typically have to make trade-offs between reliability, cost, time, and other practical considerations. In cases where such trade-offs are warranted, the following methods can increase the reliability of the results.

Randomized Field Trials

The concept of randomization in medical research is simple: Take a large group of individuals with the same characteristics and medical affliction, and randomly divide them into two subgroups.¹⁴ Administer the treatment to just one subgroup and closely monitor everyone's health. If the treated (or test) group does statistically better than the untreated (or control) group and the results can be replicated, then the therapy is deemed to be effective. Similarly, randomized field trials can help companies determine whether specific changes will lead to improved performance. The financial services company Capital One has long used randomized experiments to test even the most seemingly trivial changes. For instance, it might test the color of envelopes used for product offers by sending out two batches (one in the test color and the other in white) to random recipients and determining any differences in responses. As Capital One's cofounder and CEO Richard Fairbank explained, the same principle can be applied to more crucial issues, such as when customers call to cancel their credit cards due to a better interest rate offer from another bank:

A classic test for Capital One is to randomize all the people who are calling the retention department saying: "I'm outta here." To respond appropriately requires some knowledge of who's bluffing and who's not, as well as some knowledge about which customers we'd like to keep. To get this information, we perform a test with, for the sake of simplicity, three different actions across three randomized groups of people. Group 1, we call their bluff and close their account. Group 2, we match their (allegedly) better offer. And Group 3, we meet them halfway. Then, we collect lots of information on what the responses of these offers are, and build statistical models to link these results to the data we had on these people. So now, when somebody calls Capital One—instantaneously—we make an actuarial calculation of the customer's lifetime Net Present Value and assess the customer's likely

response. Right on the screen, the customer service rep sees an instant recommendation, such as to negotiate the APR down to 12.9 percent.¹⁵

Randomization plays an important role, as it is virtually impossible to control all variables in a business experiment. It helps prevent systemic bias, introduced consciously or unconsciously, from affecting an experiment, and it evenly spreads any remaining (and possibly unknown) potential causes of the outcome between the test and control groups. But randomized field tests are not without challenges. For the results to be valid, the field trials must be conducted in a statistically rigorous fashion, and it's easy for managers to slip.

Instead of identifying a population of test subjects with the same characteristics and then randomly dividing that population into two groups, managers sometimes make the mistake of selecting a test group (say, a group of stores in a chain) and then assuming that everything else (the remainder of the stores) should be the control group. Or they select the test and control groups in ways that inadvertently introduce biases into the experiment. Petco used to select its thirty best stores to try out a new initiative (as a test group) and compare them with its thirty worst stores (as the control group). Initiatives tested in this way would often look very promising but fail when they were rolled out. Now Petco considers a wide range of parameters—store size, customer demographics, the presence of nearby competitors, and so on—to match the characteristics of the control and test groups. (Publix does the same.) The results from those experiments have been much more trustworthy.

Blind Tests

To minimize biases and increase reliability further, Petco and Publix have conducted “blind” tests, which help prevent the so-called *Hawthorne effect*: the tendency of study participants to modify their behavior, consciously or subconsciously, when they are aware that they are part of an experiment (named after Hawthorne Works, a factory outside Chicago that conducted experiments in the early twentieth century to see if better lighting increased productivity).¹⁶ At Petco, none of the test stores' staffers know when experiments are under way; at Publix, stores are continually rolling out new prices, so the tests are indistinguishable from normal operating practices. Blind procedures make sure that experimenters and participants don't change behaviors just because they are part of a test.

Blind procedures, however, are not always practical. For tests of new equipment or work practices, Publix typically informs the stores that have been selected for the test group.

Otherwise, stores may be reluctant to participate or even confused about why the changes are made. (Note: A higher experimental standard is the use of “double-blind” tests, in which neither the experimenters nor the test subjects are aware of which participants are in the test group and which are in the control. Double-blind tests are widely used in medical research but are not commonplace in business experimentation.)

Big Data

In online and other direct-channel environments, the math required to conduct a rigorous randomized experiment is well known by data scientists, who can employ sample sizes involving millions of customers. But many consumer transactions are still conducted through complex distribution systems such as store networks, sales territories, bank branches, fast-food franchises, and so on. In such environments, sample sizes are often smaller than one hundred, violating typical assumptions of many standard statistical methods.¹⁷ To minimize the effects of this limitation, companies can utilize specialized algorithms in combination with multiple sets of big data. Interestingly, the smallest sample sizes require the most sophisticated analytical processing and big data methods. (See the sidebar “[How Big Data Can Help with Experiments.](#)”)

Consider the real example of a large retailer that was contemplating a store redesign that would cost a half-billion dollars to roll out to thirteen hundred locations. To test the idea, the retailer redesigned twenty stores and tracked the results. The finance team analyzed the data and concluded that the upgrade would increase sales by a meager 0.5 percent, resulting in a negative return on investment. The marketing team conducted a separate analysis and forecast that the redesign would lead to a healthy 5 percent sales increase.

As it turned out, the finance team had compared the test sites with other stores in the chain that were of similar size, demographic income, and other variables but were not necessarily in the same geographic market. It had also used data six months before and after the redesign. In contrast, the marketing team had compared stores within the same geographic region and had considered data twelve months before and after the redesign. To determine which results to trust, the company employed big data, including transaction-level data (store items, the times of day when the sale occurred, prices), store attributes, and data on the environments around the stores (competition, demographics, weather). In this way, the company selected stores for the control group that were a closer match with those in which the redesign was tested, which made the small sample size statistically valid. It then used objective, statistical methods to review both analyses. The results: The

marketing team's findings were the more accurate of the two, and the redesign was approved.

How Big Data Can Help with Experiments

To filter out statistical noise and identify cause-and-effect relationships, business experiments should ideally employ samples numbering in the thousands or higher. But this can be prohibitively expensive or impossible. A new approach to merchandise assortment may have to be tested in just twenty-five stores, a sales-training program with thirty-two salespeople, and a proposed remodeling in ten hotel properties. In such situations, big data and other sophisticated computing techniques, such as machine learning, can help. Here's how:

Getting Started

If a retailer wants to test a new store layout, it should collect detailed data (such as competitors' proximity, employee' tenures, and customer demography) about each unit of analysis (each store and its trade area, each salesperson and her accounts, and so on). This will become part of a big data set. Determining how many and which stores, customers, or employees should be part of the test and how long the test should run depends on the volatility in the data and the precision required for impact estimates.

Building a Control Group

In experiments involving small samples, correctly matching test subjects (such as individual stores or customers) to control subjects is essential and depends on the experimenter's ability to fully identify dozens or even hundreds of variables that characterize the test subjects. Big data feeds (complete transaction logs by customer, detailed weather data, social media streams, and so on) can assist in this. Once the characteristics are determined, a control group can be built that contains all elements of the test group except for what is being tested. This allows the retailer to determine whether the test results were influenced only by that one element—the new layout—or by other factors (demographic variances, better economic conditions, warmer weather).

Targeting the Best Opportunity

The same data feeds can be used to identify situations in which the tested program is effective. For example, the new store layout may work better in highly competitive urban areas but may be only moderately successful in other markets. By pinpointing these patterns, the experimenter can implement the program in situations where it works and avoid investments where it may not generate the best return on investment (ROI).

Tailoring the Program

Additional large data feeds can be used to characterize program components that are more effective or less effective. For example, a retailer testing the effects of a new store layout can use data captured from in-store video streams to determine whether the new layout is encouraging customers to move through more of the store or is generating more traffic near high-margin products. Or the experimenter may find that moving items to the front of the store and putting in new shelves have a positive impact, but moving the sales registers disrupts checkouts and hurts profits.

Even when a company can't follow a rigorous testing protocol, analysts can help identify and correct for certain biases, randomization failures, and other experimental imperfections. In one common situation, an organization's testing function is presented with nonrandomized natural experiments—the vice president of operations, for example, might want to know if the company's new employee training program, which was introduced in about 10 percent of the company's markets, is more effective than the old one. As it turns out, in such situations the same algorithms and big data sets that can be used to address the problem of small or correlated samples can also be deployed to tease out valuable insights and minimize uncertainty in the results. The analysis can then help experimenters design a true randomized field trial to confirm and refine the results, especially when they are somewhat counterintuitive or are needed to inform a decision with large economic stakes.

For any experiment, the gold standard is *replication*; that is, others conducting the same test should obtain similar results. Repeating an expensive test is usually impractical, but

companies can verify results in other ways. Petco sometimes deploys a staged rollout for large initiatives to confirm the results before proceeding with a companywide implementation. And Publix has a process for tracking the results of a rollout and comparing them with the predicted benefit.

Question 5: Do We Understand Cause and Effect?

Because of the excitement over big data, some executives may mistakenly believe that causality isn't important and experimental controls are optional. In their minds, all they need to do is establish correlation, and causality can be inferred. But it's not that simple. Sometimes two variables are correlated because they have a common cause, such as the relationship between drowning and ice cream consumption (outside temperature), or because the correlation is simply a coincidence. In one analysis, researchers found that the number of lawyers in California correlates very highly with money spent on pets in the United States.¹⁸ I leave plausible explanations to your imagination.

To categorize the different levels of understanding causality, Judea Pearl and Dana Mackenzie propose a three-tier ladder in *The Book of Why*.¹⁹ The first and lowest causality tier, *association*, is about finding regularities in observations. One event is associated, or correlated, with another if observing one changes the likelihood of observing the other. The authors place modern-day analytics and big data in this tier. The second, *intervention*, requires changing one or more variables and observing changes in outcomes. Experiments are such interventions. The third and highest tier, *counterfactuals*, includes the strongest test of causality. Instead of just asking, "Did A cause B?" a higher standard includes the counterfactual, "Would B have occurred if not for A?" Here is an example from my teenage years: I had a friend who firmly believed that drinking a glass of salt water (A) after alcohol consumption prevented a hangover (B) the next day. But would his hangover (B) have occurred if he didn't drink salt water? The difficulty with counterfactuals is that you can't go back in time, repeat the experiment with a different or no intervention, and then compare two outcomes using the same person. (Often, personal remedies are based on anecdotes or a handful of personal experiences that could have been explained by other factors; in my friend's case, I suspect that the thought of drinking nauseating salt water may

have compelled him to consume less alcohol.) In [chapter 3](#), we will learn how causal effects are “estimated” through randomized, controlled online experiments.²⁰

Notwithstanding these causality tiers, the excitement over big data led to the extraordinary claim that the scientific method is no longer needed. In 2008, *Wired* magazine published a provocative article, “The End of Theory: The Data Deluge Makes the Scientific Method Obsolete,” using Google as an example of an organization that succeeded without any models of cause and effect.²¹ Similarly, books on big data cited anecdotes in which correlation was now sufficient to make important business decisions.²² One popular company example was—once again—Google and how its Flu Trend algorithms simply mined five years of web logs containing hundreds of billions of searches and thus became better at predicting the incidence of flu than government statistics. But in 2014, a team of Harvard-affiliated researchers found that between August 21, 2001, and September 1, 2013, Google’s algorithms had overestimated flu prevalence in 100 out of 108 weeks!²³ Importantly, many publications overlooked the fact that Google isn’t just a big data miner but also a ferocious experimenter. The company understands that correlations are excellent sources of hypotheses that need to be rigorously tested for causality. The Mark Twain adage comes to mind: “The rumors of [the scientific method’s] death have been greatly exaggerated.”

The following two examples further illustrate the difficulty of inferring causality from correlation—and also highlight the shortcomings of experiments that lack control groups.²⁴ The first concerns two teams that conducted separate observational studies of two advanced features for Microsoft Office. Each concluded that the new feature it was assessing reduced attrition. In fact, almost any advanced feature will show such a correlation, because people who will try an advanced feature tend to be heavy users, and heavy users tend to have lower attrition. So while a new advanced feature might be correlated with lower attrition, it doesn’t necessarily cause it. Office users who get error messages also have lower attrition because they too tend to be heavy users. But does that mean that showing users more error messages will reduce attrition? Hardly.

The second example concerns a study Yahoo did to assess whether display ads for a brand shown on Yahoo sites could increase searches for the brand name or related keywords. The observational part of the study estimated that the ads increased the number of searches by 871 percent to 1,198 percent. But when Yahoo ran a controlled experiment, the increase was only 5.4 percent. If not for the control, the company might have concluded that the ads

had a huge impact and wouldn't have realized that the increase in searches was due to other variables that changed during the observation period.

Clearly, observational studies cannot establish causality. This is well known in medicine, which is why the US Food and Drug Administration mandates that companies conduct randomized, controlled clinical trials to prove that their drugs are safe and effective. Of course, there are circumstances when controlled experiments are neither practical nor ethical. In such cases, great care must be taken to remove, investigate, and measure bias in observational studies.²⁵ But it's important to remain skeptical when drawing conclusions from nonrandomized studies. In a famous study of forty-five highly cited clinical research studies on the effectiveness of medical interventions (e.g., therapies, procedures, medicine), only 17 percent of nonrandomized studies stood up to replication by subsequent studies that had a stronger research design. By contrast, 77 percent of the findings of randomized studies were replicated.²⁶

Just understanding simple cause and effect isn't always enough. What if you are able to determine that one thing causes another, but you don't know why? Should you try to understand the causal mechanism? The short answer is yes, particularly when the stakes are high. Between 1500 and 1800, an estimated 2 million sailors died of scurvy.²⁷ Today we know that scurvy was caused by lack of vitamin C in the diet of sailors, who didn't have adequate supplies of fruit on long voyages. In 1747, Dr. James Lind, a surgeon in the Royal Navy, decided to conduct an experiment to test six possible cures. On one voyage he gave some sailors oranges and lemons, and others alternative remedies like vinegar. The experiment showed that citrus fruits could prevent scurvy, although no one knew why. Lind mistakenly believed that the acidity of the fruit was the cure and tried to create a less perishable remedy by heating the citrus juice into a concentrate, which destroyed the vitamin C. It wasn't until fifty years later, when unheated lemon juice was added to sailors' daily rations, that the Royal Navy finally eliminated scurvy among its crews. Presumably, the cure could have come much earlier and saved many lives if Lind had run a controlled experiment with heated and unheated lemon juice. In the same way, companies may be better at implementing changes effectively if they know why the change has the effect they desire, so they aren't implementing the change in the wrong way or wasting resources on elements that don't matter.

That said, we don't always have to know the "why" or "how" to benefit from knowledge of the "what." This is particularly true when it comes to the behavior of users, whose motivations can be difficult to determine. At Microsoft's Bing, some of the biggest

breakthroughs were made without an underlying theory. In 2013, for example, Bing ran a set of experiments with the colors of various text that appeared on its search-results page, including titles, links, and captions ([chapter 3](#) has more details).²⁸ Though the color changes were subtle, the results were unexpectedly positive: they showed that users who saw the slightly darker shades in titles and slightly lighter shades in captions were successful in their searches a larger percentage of the time and found what they wanted in significantly less time. Even though Bing was able to improve the user experience with those subtle changes in type colors, the company had no well-established theories about color that could help it understand why. Here, the evidence and rigor of experimental protocols created trust in the results and took the place of theory.

A similar phenomenon happened at Petco. When executives investigated new pricing for a product sold by weight, the results were unequivocal. By far, the best price was for a quarter pound of the product, and that price was for an amount that ended in \$.25. That result went sharply against the grain of conventional wisdom, which typically calls for prices ending in 9, such as \$4.99 or \$2.49. “This broke a rule in retailing that you can’t have an ‘ugly’ price,” notes Rhoades. At first, executives at Petco were skeptical of the results, but because the experiment had been conducted so rigorously, they eventually were willing to give the new pricing a try. A targeted rollout confirmed the results, leading to a sales jump of more than 24 percent after six months.

But without fully understanding causality, companies leave themselves open to making big mistakes. Remember the experiment Kohl’s conducted to investigate the effects of delaying the opening of its stores? During that testing, the company suffered an initial drop in sales. At that point, executives could have pulled the plug on the initiative. But an analysis showed that the number of customer transactions had remained the same; the issue was a drop in units per transaction. Eventually, the units per transaction recovered and total sales returned to previous levels. Kohl’s couldn’t fully explain the initial decrease, but executives resisted the temptation to blame the reduced operating hours. They didn’t rush to equate correlation with causation.

Question 6: Have We Gotten the Most Value Out of the Experiment?

Many companies go through the expense of conducting experiments but then fail to make the most of them. To avoid that mistake, executives should take into account a proposed initiative’s effect on various customers, markets, and segments and concentrate

investments in areas where the potential paybacks are highest. The best question is usually not “What works?” but “What works where?” or “What is surprising?”

Petco frequently rolls out a program only in stores that are most similar to the test stores that had the best results. By doing so, it not only saves on implementation costs but also avoids involving stores where the new program might not deliver benefits or might even have negative consequences. Thanks to such targeted rollouts, Petco has consistently been able to double the predicted benefits of new initiatives.

Another useful tactic is *value engineering*. Most programs have some components that create benefits in excess of costs and others that do not. The trick, then, is to implement just the components with an attractive return on investment (ROI). As a simple example, let’s say that a retailer’s tests of a 20-percent-off promotion show a 5 percent lift in sales. What portion of that increase was due to the offer itself, and what resulted from the accompanying advertising and training of store staff, both of which directed customers to those particular products? In such cases, companies can conduct experiments to investigate various combinations of components (for instance, the promotional offer with advertising but without additional staff training). An analysis of the results can disentangle the effects, allowing executives to drop the components (say, the additional staff training) that have a low or negative ROI.

Moreover, a careful analysis of data generated by experiments can enable companies to better understand their operations and test their assumptions of which variables cause which effects. With big data, the emphasis is on correlation—discovering, for instance, that sales of certain products tend to coincide with sales of others. But business experimentation can allow companies to look beyond correlation and investigate causality—uncovering, for instance, the factors causing the increase (or decrease) of purchases. Such fundamental knowledge of causality can be crucial. Without it, executives have only a fragmentary understanding of their businesses, and the decisions they make can easily backfire.

When Cracker Barrel Old Country Store, the Southern-themed restaurant chain, conducted an experiment to determine whether it should switch from incandescent to LED lights at its restaurants, executives were astonished to learn that customer traffic actually decreased in the locations that installed LED lights. The lighting initiative could have stopped there, but the company dug deeper to understand the underlying causes. As it turned out, the new lighting made the front porches of the restaurants look dimmer, and many customers mistakenly thought that the restaurants were closed. This was puzzling—

the LEDs should have made the porches brighter. Upon further investigation, executives learned that the store managers hadn't previously been following the company's lighting standards; they had been making their own adjustments, often adding extra lighting on the front porches. Thus the luminosity dropped when the stores adhered to the new LED policy. The point here is that correlation alone would have left the company with the wrong impression—that LEDs are bad for business. It took experimentation to uncover the actual causal relationship.

What's important here is that many companies are discovering that conducting an experiment is just the beginning. Value comes from analyzing and then exploiting the data. In the past, Publix spent 80 percent of its testing time gathering data and 20 percent analyzing it. The company's current goal is to reverse that ratio.

Question 7: Are Experiments *Really* Driving Our Decisions?

Not all management decisions can or should be resolved by experiments. The decisions to acquire another company or to enter a new market segment are best left to judgment, observation, and analysis. Sometimes running an experiment can be very difficult, if not impossible, or imposes so many constraints on the experimenter that the results aren't useful. But if everything that can be tested *is* tested, experiments can become instrumental to management decision-making and fuel healthy debates. That's what happened at Netflix, which has built a sophisticated infrastructure for large-scale experimentation. According to the *Wall Street Journal*, the company's executives were torn when tests showed that a promotional image that included *only* Lily Tomlin, one of the stars of the comedy *Grace and Frankie* in 2016, resulted in more clicks by potential viewers than images showing both Tomlin and her costar Jane Fonda.²⁹ The content team was concerned that excluding Fonda would alienate the actor and possibly violate her contract. After heated debates that pitted empirical evidence against "strategic considerations," Netflix chose to use images that also included Fonda, even though customer data didn't support the decision. However, the experimental evidence made the trade-offs and decision-making process more transparent.

When they do choose to make decisions according to test findings, companies should ensure the validity of their test results by paying attention to sample sizes, control groups, randomization, and other factors. The more valid and repeatable the results, the better they will hold up in the face of internal resistance, which can be especially strong when results challenge long-standing industry practices and assumptions. More importantly, hierarchy

and PowerPoint presentations should not be accepted as a substitute for experimental evidence.

Consider how running business experiments changed decision-making at Bank of America when the company studied waiting time in bank branches.³⁰ Around the year 2000, the bank operated some forty-five hundred banking centers in twenty-one states, serving approximately 27 million households and 2 million businesses, and processing 3.8 million transactions each day. Internal researchers, who “intercepted” some thousand customer at bank lines, noted that after about three minutes, the gap between actual and perceived wait time rose exponentially. Two focus groups with sales associates and a formal analysis by the Gallup organization provided further corroboration—and the transaction zone media (TZM) experiment was born. The team speculated, based on published psychology literature, that “entertaining” clients through television monitors above the lobby tellers would reduce perceived wait times by at least 15 percent. The team chose two similar branches for the TZM experiment and its control so they could maximize their learning from the experiment. It installed monitors set to the Atlanta-based news station CNN over teller booths in the branch. The team then waited for a week’s “washout” period to allow the novelty to wear off before measuring results for the subsequent two weeks.

Results from the TZM-equipped branch showed that the number of people who overestimated their actual wait times dropped from 32 percent to 15 percent. During the same period, none of the other branches reported drops of this magnitude. In fact, the branch used as a control saw an increase in overestimated wait times from 15 percent to 26 percent. Though these were encouraging results, the team still had to prove to senior management that the TZM could positively affect the corporate bottom line. To do so, the team relied on a model that used the easily measurable “Customer Satisfaction Index” that they created (based on a thirty-question survey) as a proxy for future revenue growth.

Prior studies indicated that every one-point improvement in the Customer Satisfaction Index corresponded to a \$1.40 in added annual revenue per household from increased customer purchases and retention. Thus, an increase in the index of just two points for a banking center (branch) with a customer base of ten thousand households would increase its annual revenues by \$28,000. Percentages generally ranged in the mid-80s in Atlanta, Bank of America’s test market, and in the high 70s to low 80s nationally. The team measured an overall 1.7 percent increase after installation of the TZM monitors. Sufficiently encouraged, they entered a second phase to study and optimize the impact of more varied programming, advertising, and different sound speaker parameters.

While the benefits of the TZM program were laudable, the team now had to consider whether they outweighed the costs. Studies indicated that it would cost some \$22,000 to install the special TV monitors at each branch that was part of its experimentation portfolio. For a national roll-out, the estimated economies of scale would bring costs down to about \$10,000 per site, which could be directly compared against the implied financial benefit for the bank.

The lesson from Bank of America, Kohl's, Publix, and the other examples in this chapter is not merely that business experimentation can lead to better ways of doing things if managers ask the right questions (see [table 2-1](#)). Experimentation can also help overturn wrongheaded conventional wisdom and the faulty business intuition of even seasoned executives. And smarter decision making will ultimately lead to improved performance. In general, better testing has helped save Publix tens of millions of dollars by doing two things: First, it provides the company with the confidence to proceed with innovative proposals that will improve performance. And second, it helps the company avoid making a change that could ultimately damage the bottom line.

Could the disaster at J.C. Penney described in [chapter 1](#) have been averted by testing the various changes beforehand (for example, the addition of branded boutiques)? At this point, it's impossible to know. But one thing's for sure: before attempting to implement such a bold program, the company needed more experimental evidence and less executive intuition to guide those decisions.